Disproof of Conjecture 00000002160: the Higman–Sims graph does not attain the Hoffman bound

Submission `aristotle_submission_20261020150000`

2026-10-20

1 The conjecture

Conjecture 00000002160 reads:

> *Definition: The independence number of the Higman–Sims graph. Conjecture: The independence number equals the Hoffman bound, and the kernel of the bound is the Delsarte semilattice.*

The statement is a conjunction. We refute its first, precise, conjunct; hence the whole conjunction is false. (The second conjunct, "the kernel of the bound is the Delsarte semilattice", has no standard mathematical meaning and is not used.)

2 Definitions

Definition 1 (Hoffman bound)**.** Let G be a k-regular graph on n vertices whose adjacency matrix A has least eigenvalue $\lambda_{\min}$. Hoffman's ratio bound is

$$H(G) \;=\; \frac{n\,(-\lambda_{\min})}{k-\lambda_{\min}}.$$

Hoffman's theorem states $\alpha(G) \le H(G)$, where α is the independence number; "$\alpha(G)$ equals the Hoffman bound" (equivalently, "the Hoffman bound is attained") means $\alpha(G) = H(G)$.

Definition 2 (Higman–Sims graph)**.** Let $\mathcal{S}$ be the Steiner system $S(3,6,22)$: 22 points and 77 six-element blocks (hexads) such that any 3 points lie in exactly one hexad. The Higman–Sims graph HS has the 100 vertices $\{\infty\} \cup \{\text{points}\} \cup \{\text{hexads}\}$; ∞ is adjacent to every point, a point is adjacent to the hexads containing it, and two hexads are adjacent iff they are disjoint. It is the (unique) strongly regular graph with parameters $(100, 22, 0, 6)$.

3 The disproof

Theorem 3. *HS is 22-regular, its least adjacency eigenvalue is -8, and $H(HS) = \frac{100\cdot 8}{22+8} = \frac{80}{3}$. Consequently $\alpha(HS) \neq H(HS)$, i.e. the conjecture is false.*

Lemma 4. *Let G be any strongly regular graph with parameters $(100, 22, 0, 6)$ and adjacency matrix A. Then every eigenvalue of A lies in $\{22, 2, -8\}$, and -8 is an eigenvalue.*

Proof. Strong regularity says $A^2 = 22I + 0 \cdot A + 6(J - I - A)$, where J is the all-ones matrix. Let $Av = \mu v$ with $v \neq 0$ and put $s = \sum_x v_x$. Applying the identity to v gives $(\mu^2 + 6\mu - 16)\, v = 6s\, \mathbf{1}$. If $\mu^2 + 6\mu - 16 \neq 0$, then v is a nonzero constant vector, and since G is 22-regular, $Av = 22v$, so $\mu = 22$. Otherwise $(\mu - 2)(\mu + 8) = 0$. Hence every eigenvalue is 22, 2 or -8.

The eigenvalues $\mu_1, \ldots, \mu_{100}$ (with multiplicity) sum to $\operatorname{tr} A = 0$. If none of them were -8, each would be ≥ 2 and their sum would be $\geq 200 > 0$, a contradiction. □

Proof of Theorem **??**. By Lemma **??**, $\lambda_{\min} = -8$, and $k = 22$, $n = 100$, so $H(HS) = 800/30 = 80/3$. The independence number is a natural number, and $3\alpha = 80$ has no integer solution; hence $\alpha(HS) \neq 80/3$. □

Remark 5. The neighbourhood of ∞ (the 22 points) is independent, so $\alpha(HS) \geq 22$; this is also checked in Lean. It is known that $\alpha(HS) = 22$; this fact is *not* needed and not formalised here. Under the alternative reading "$\alpha(HS) = \lfloor H(HS) \rfloor = 26$" the conjecture would also be false by this known value, but that reading is not covered by the formal proof.

4 Lean formalisation

The Lean 4 project in `lean/` (Lean 4.28.0, Mathlib `v4.28.0`) contains:

- `Conjecture2160/HigmanSims.lean`: the 77 hexads of $S(3, 6, 22)$ as 22-bit masks (derived from the extended binary Golay code), a kernel-checked proof (`hexads_steiner`) that they form an $S(3, 6, 22)$, the graph `higmanSims : SimpleGraph (Fin 100)` defined exactly as above (vertex $0 = \infty$, $1, \ldots, 22$ = points, $23, \ldots, 99$ = hexads), and `higmanSims_isSRG`: `higmanSims.IsSRGWith 100 22 0 6` (Mathlib's notion). The latter is proved by `decide +kernel` on a Boolean checker together with a soundness proof linking the checker to Mathlib's `commonNeighbors` and `degree`.
- `Conjecture2160/Spectral.lean`: `minAdjEigenvalue` (least eigenvalue of the real symmetric adjacency matrix, via Mathlib's spectral theorem), `hoffmanBound`, and Lemma **??** for every $\mathrm{srg}(100, 22, 0, 6)$ (`srg_minAdjEigenvalue`, `srg_hoffmanBound`, `srg_indepNum_ne_hoffmanBound`).
- `Conjecture2160/Main.lean`: the main theorem `conjecture_2160_false`, stating `higmanSims.IsRegularOf 22` and (`higmanSims.indepNum` : $\mathbb{R}$) $\neq$ `hoffmanBound higmanSims 22`, and `higmanSims_hoffmanBound : hoffmanBound higmanSims 22 = 80/3`.

All theorems depend only on the axioms `propext`, `Classical.choice`, `Quot.sound`; there is no `sorry` and no `native_decide`. The auxiliary script `gen_hs.py` regenerates the hexad and row data and independently re-checks the Steiner and strong-regularity properties.